CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

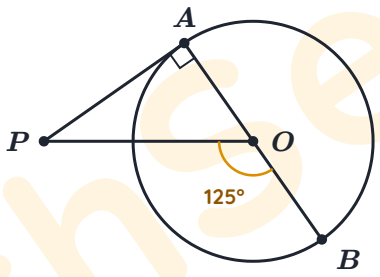
Chapter 10: Circles

Previous year board questions, topic-wise: 27 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Theorem 1: tangent and radius

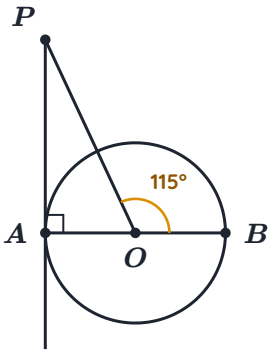
1. PA is a tangent from an external point P to a circle with centre O, touching it at A. AOB is a diameter and $\angle POB = 125^\circ$. Find $\angle APO$. (a) 25° (b) 65° (c) 90° (d) 35° CBSE 2026, 1 mark



2. Two concentric circles have radii 5 cm and 4 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. CBSE 2026, 2 marks

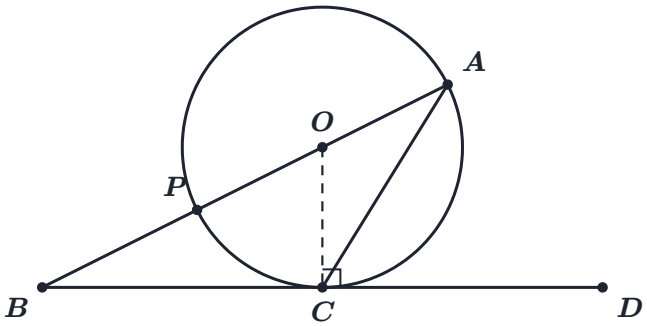
3. Tangents are drawn to a circle at the two ends of a diameter. How are the two tangents related? CBSE 2025, 1 mark

4. AOB is a diameter of a circle with centre O, and PA is the tangent at A. If $\angle POB = 115^\circ$, find $\angle APO$. CBSE 2025, 1 mark



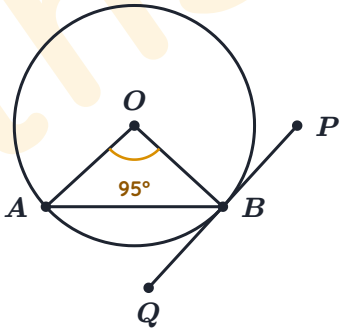
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5. O is the centre of a circle and BCD is the tangent to it at C. The line from B through O meets the circle at P and A. Prove that $\angle BAC + \angle ACD = 90°$. CBSE 2025, 2 marks

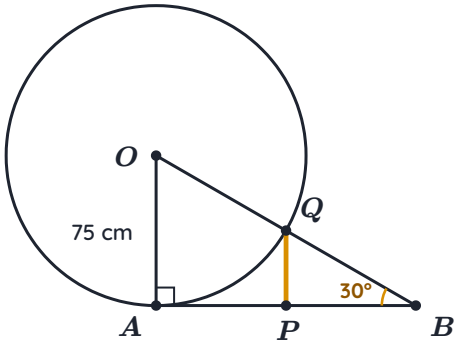


6. Assertion (A): the tangents at the two ends of a diameter of a circle are parallel. Reason (R): the diameter is the longest chord of a circle. Choose: (a) A and R are both true and R explains A. (b) A and R are both true but R does not explain A. (c) A is true and R is false. (d) A is false and R is true. CBSE 2024, 1 mark

7. PQ is the tangent at B to a circle with centre O, and AB is a chord with $\angle AOB = 95°$. Q lies on the same side of OB as A. Find $\angle ABQ$. (a) $47.5°$ (b) $42.5°$ (c) $85°$ (d) $95°$ CBSE 2023, 1 mark

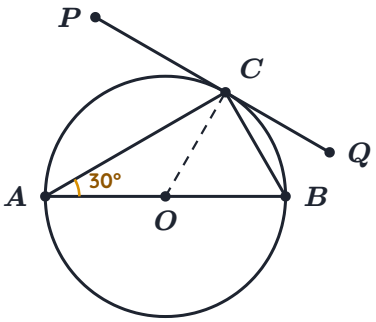


8. A discus leaves the thrower's circular path along a tangent. AB is such a tangent, touching a circle of radius 75 cm and centre O at A, with $\angle ABO = 30°$. OB cuts the circle at Q, and P is the point on AB with $PQ \parallel OA$. (i) Find AB. (ii) Find OB. (iii) Find AP, OR find PQ. CBSE 2023, 4 marks

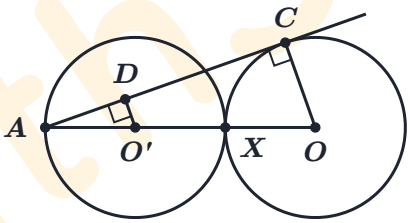


9. Two concentric circles have radii 4 cm and 3 cm. Find the length of the chord of the larger circle that touches the smaller circle. CBSE 2022 Term 2, 2 marks

10. PQ is the tangent at C to a circle with centre O, and AB is a diameter with $\angle CAB = 30°$. P lies on the same side of OC as A. Find $\angle PCA$. CBSE 2016, 1 mark



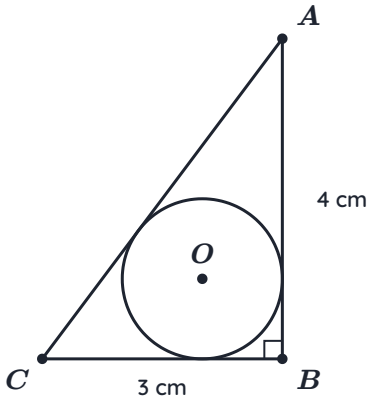
11. Two equal circles with centres O and O' touch each other at X. The line OO' produced meets the circle with centre O' again at A. AC is a tangent from A to the circle with centre O, touching it at C, and $O'D \perp AC$. Find $\frac{DO'}{CO}$. CBSE 2016, 4 marks



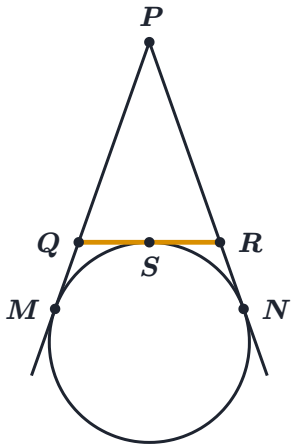
2 Theorem 2: tangents from an external point

12. TP and TQ are tangents from an external point T to a circle with centre O, and $\angle POQ = 120°$. Find $\angle PTQ$. (a) $60°$ (b) $70°$ (c) $80°$ (d) $90°$ CBSE 2026, 1 mark

13. $\triangle ABC$ is right-angled at B with $AB = 4$ cm and $BC = 3$ cm. Find the radius of the circle inscribed in the triangle. CBSE 2026, 3 marks

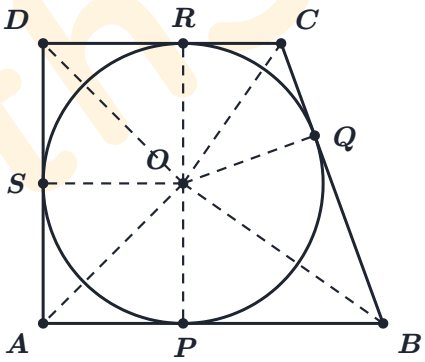


14. A circle touches the side QR of $\triangle PQR$ at S, and touches PQ and PR produced at M and N. Prove that $PM = \frac{1}{2}(PQ + QR + PR)$. CBSE 2026, 3 marks



15. A point P is 26 m from the centre of a circle. Each tangent from P to the circle is 10 m long. Find the radius of the circle. CBSE 2025, 2 marks

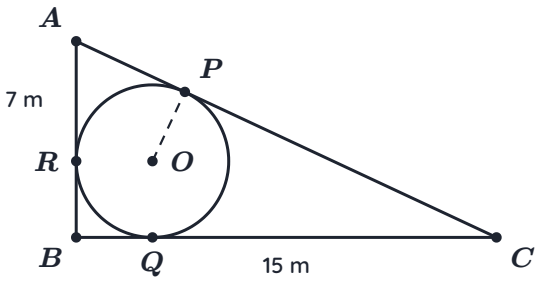
16. A quadrilateral is drawn around a circle so that all four sides touch it. Prove that each pair of opposite sides subtends supplementary angles at the centre. CBSE 2025, 2 marks



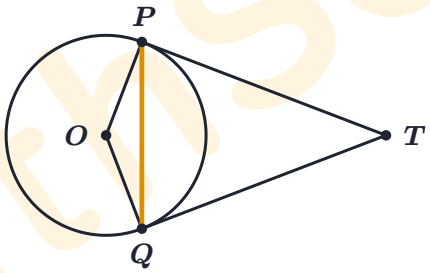
17. AB is a diameter of a circle with centre O. The tangents at A and B are met by a third tangent at Q and P respectively. Prove that $\angle POQ = 90°$. CBSE 2024, 3 marks

18. A circle of radius 8 cm with centre O touches the sides AD, DC, CB and BA of a quadrilateral $ABCD$ at P, Q, R and S. $AD \perp DC$, $BC = 30$ cm and $BS = 24$ cm. Find DC. CBSE 2024, 3 marks

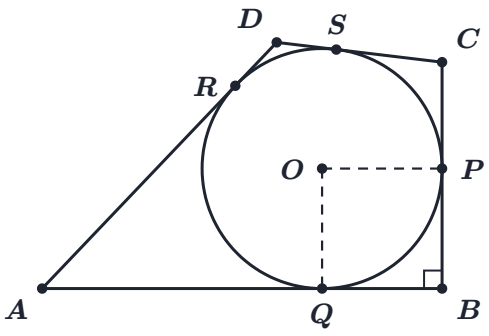
19. A triangular backyard ABC has a right angle at B, with $AB = 7$ m and $BC = 15$ m. A circular pit inside it touches AC, BC and AB at P, Q and R, and $AP = x$ m. (i) Write AR in terms of x. (ii) What kind of quadrilateral is $BQOR$? (iii) Write PC in terms of x and find x, OR find x and then the radius r of the pit. CBSE 2024, 4 marks, case study



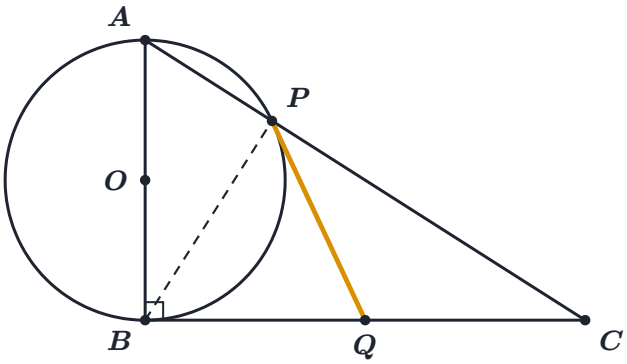
20. TP and TQ are tangents from an external point T to a circle with centre O. Prove that $\angle PTQ = 2\angle OPQ$. CBSE 2023, 3 marks



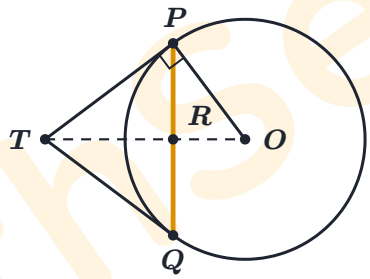
21. A circle with centre O is inscribed in a quadrilateral $ABCD$, touching AB, BC, CD and DA at Q, P, S and R. $\angle B = 90°$, $AD = 17$ cm, $AB = 20$ cm and $DS = 3$ cm. Find the radius of the circle. CBSE 2023, 3 marks



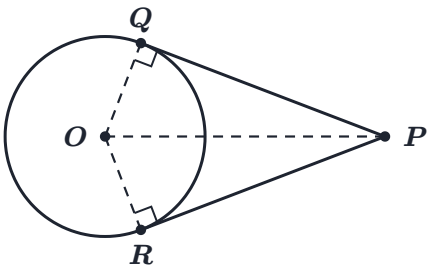
22. $\triangle ABC$ is right-angled at B. A circle drawn on AB as diameter cuts AC at P. The tangent to the circle at P meets BC at Q. Prove that Q is the mid-point of BC. CBSE 2022 Term 2, 4 marks



23. PQ is a chord of length 8 cm in a circle of radius 5 cm with centre O. The tangents at P and Q meet at T. Find TP. CBSE 2019, 3 marks

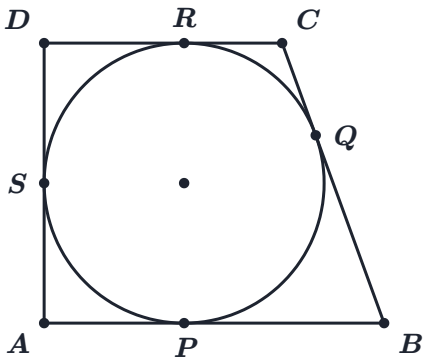


24. Prove that the two tangents drawn to a circle from a point outside it are equal in length.
CBSE 2018, 3 marks

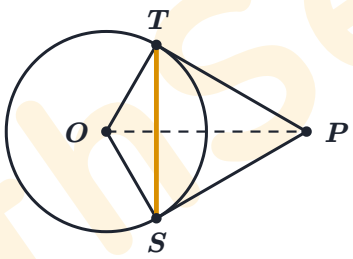


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25. A quadrilateral $ABCD$ is drawn around a circle, with AB, BC, CD and DA touching the circle at P, Q, R and S. Prove that $AB + CD = BC + DA$. CBSE 2016, 2 marks



26. PT and PS are tangents from an external point P to a circle with centre O and radius r. If $OP = 2r$, show that $\angle OTS = \angle OST = 30°$. CBSE 2016, 2 marks



27. Prove that the tangents drawn to a circle from an external point are equal in length.
CBSE 2016, 4 marks

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